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Received: 18 September 2022 | Revised: 27 June 2023 | Accepted: 3 July 2023

DOI: 10.1111/psyp.14390

PSYCHOPHYSIOLOGY SPR WILEY

ORIGINAL ARTICLE

The effect of choice on memory: The role of theta oscillations

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Funding information
Natural Science Foundation of China, Grant/Award Number: 31500888 and 62177045; German Research Foundation, Grant/Award Number: TRR-169; National Science Foundation of China, Grant/Award Number: 61621136008

Abstract

People value the opportunity to exercise control over the environment or make their own choices. Recent studies have revealed that simply having the opportunity to make choices can facilitate memory performance, suggesting an interaction between reward (due to choice making) and memory systems. However, little is known about the electrophysiological basis of choice-related memory. In the current study, we used scalp electroencephalography combined with a choice encoding task to examine the role of theta oscillations (which have been widely connected to reward and memory processing) in choice-related memory formation. The encoding task had two conditions. In the choice condition, participants were asked to choose between two occluded memoranda by themselves, whereas in the fixed condition, the decision was made by the computer. Behavioral results showed the choice effect, with better performance in the choice condition than the fixed condition on the recognition test given after a 24-h delay. Increases in theta power during an early latency of encoding period predicted successful memory formation in the choice condition, but not in the fixed condition. Furthermore, decreases in theta power during a late latency predicted successful memory formation in both the fixed and the choice conditions. Finally, we observed increased theta power in the choice condition compared to the fixed condition during an early latency of encoding period and decreased theta power in the choice condition compared to the fixed condition during a late latency. Our results suggest that theta oscillations play a significant role in choice-related memory formation.

KEYWORDS

choice-making, EEG, memory, motivation, subsequent memory effect (SME), theta oscillations

1 | INTRODUCTION

Human beings value the opportunity to exercise control over the environment or determine their own actions (Leotti et al., 2010; Leotti & Delgado, 2011). When individuals can determine what and how to learn and memorize, they are more motivated and their performance improves (Leotti & Delgado, 2011). This enhancement effect due

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to choice-making or self-determination occurs not only when the choices involve some aspects of the memory task (e.g., items' presentation order or duration) (Markant DuBrow et al., 2014; Murty et al., 2015; Voss et al., 2011), but also when the choices are not related to the content or process of the memory task (Ding et al., 2021). For example, in several previous studies (Ding et al., 2021; Murty et al., 2015, 2019), participants were asked to remove one of two occluder screens to intentionally remember the object underneath. In the choice condition, participants could decide which occluder to remove, whereas in the fixed condition, they were forced to remove one occluder. In this paradigm, the memoranda of each trial were predetermined and matched in the two conditions. Results showed that participants had better recognition memory in the choice condition than in the fixed condition, suggesting that the choice itself is rewarding and enhances memory performance.

Functional MRI studies have found that choice-making activates the reward system of the brain (e.g., the striatum) (Ariely & Norton, 2008; Fujiwara et al., 2013; Leotti et al., 2010), just as would other rewards such as money (Miendlarzewska et al., 2016; Murty & Adcock, 2017). The reward system is believed to interact with the hippocampus memory system to produce choice-induced memory enhancement (Murty et al., 2015, 2019). However, no study has investigated the electrophysiological basis of choice-related memory. Using scalp electroencephalography (EEG), this study focused on the relationship between theta band oscillations (4–7 Hz) and choice-related memory.

We focused on theta band oscillations because two lines of research have linked them to choice-related memory. First, theta oscillations are critical for memory. Previous studies of small mammals have linked theta rhythm to memory formation (Buzsáki & Moser, 2013; Hasselmo, 2005). Consistently, human studies also revealed that theta oscillations during the encoding (or even before encoding) of an item could determine whether it would be remembered later. This so-called subsequent memory effect (SME) in theta band or encoding/memory-related theta oscillations (Paller & Wagner, 2002) have been characterized by using scalp EEG (Klimesch et al., 1997; Paller et al., 1987; Sederberg et al., 2006), magnetoencephalography (MEG) (Guderian et al., 2009; Osipova et al., 2006), and intracranial EEG (iEEG) (Fell et al., 2001; Fernández et al., 1999; Sederberg et al., 2003).

Theta frequency has exhibited both increased and decreased oscillatory power during successful memory formation (called positive and negative SMEs, respectively) (Burgess & Gruzelier, 1997; Guderian et al., 2009; Hanslmayr & Staudigl, 2014; Klimesch, 1999; Lega et al., 2012; Osipova et al., 2006; Sederberg et al., 2003; Sederberg et al., 2006). Most previous studies have found that theta power during or even before encoding was positively correlated with subsequent memory (Fell et al., 2011; Guderian et al., 2009; Hanslmayr et al., 2009; Hanslmayr et al., 2011; Khader et al., 2010; Osipova et al., 2006; Rutishauser et al., 2010). The increased theta activity was always observed in frontal mid-line and was suggested to index intentional activation of a mnemonic context (Guderian et al., 2009) or active control of attention and cognitive effort (Bosseler et al., 2013; Nyhus & Curran, 2010), which led to better memory. Some recent studies also reported negative SMEs in the theta band (Burke et al., 2013; Greenberg et al., 2015; Guderian et al., 2009; Long et al., 2014; Sederberg et al., 2007). The negative theta SME was found in major memory-related regions such as the hippocampus and left prefrontal cortex (Greenberg et al., 2015; Long et al., 2014). It may reflect either reduced neuronal correlations between the hippocampus and the cortex, enabling the network to process and store more information (Greenberg et al., 2015), or increased neural coding capacity, leading to strengthened representation of information content (Hanslmayr et al., 2016; Staudigl & Hanslmayr, 2013).

The second line of evidence directly connects theta oscillations to the processing of reward and motivation and its role in memory and learning. Studies of rodents, monkeys, and humans have found that theta band frequency is involved in the reward (food and money) and motivation processing, such as reward expectation and exploration in reward-learning tasks (Cavanagh et al., 2012; Jutras et al., 2013; van Wingerden et al., 2010). First, in rodent study, van Wingerden et al. (2010) found that in olfactory discrimination learning, theta oscillations were associated with anticipation and delivery of reward after reward learning. Second, in a monkey study, Jutras et al. (2013) found that when monkeys were intrinsically motivated to explore images in a visual exploration task, the theta power was enhanced, resulting in superior memory of the images. Two subsequent human studies also tested the relationship between monetary reward, theta activity, and memory encoding in humans (Gruber et al., 2013; Pu & Yu, 2019). In the study by Gruber and his colleagues, participants performed an intentional memory task in high and low monetary reward conditions. Results showed that the monetary reward facilitated memory performance and that theta activity specifically predicted memory performance in the high-reward condition (Gruber et al., 2013). Pu and Yu (2019) used an incidental memory task in which the monetary reward accompanied some of the items during the encoding phase, and found that theta activity was related to memory performance for the rewarded items. These results implicate theta oscillations in

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reward/value processing and memory formation (Gruber et al., 2013).

As the brief literature review above shows, theta oscillations play an important role in reward and motivation processing and memory formation, but no study has investigated whether and how theta oscillations are associated with the formation of choice-related memory. In this study, we tested the hypothesis that theta oscillations would be associated with choice-related memory performance in humans. EEG data were recorded when participants were performing a choice encoding task, which has been used in previous studies (Ding et al., 2021; Murty et al., 2015, 2019; Xue et al., 2023). A subsequent behavioral memory test was conducted 24 h later. We were mainly interested in whether and how theta oscillations during encoding contributed to choice-related memory. Because previous studies (Gruber et al., 2013; Guderian et al., 2009) also revealed that pre-encoding theta activity was related to memory performance, we also explored theta oscillations during the pre-encoding period.

2 | METHOD

2.1 | Participants

Thirty-four healthy right-handed volunteers took part in the study. Nine participants were excluded for the following reasons: insufficient artifact-free trials (fewer than 12 trials, for details, see the Analyses section) for any condition ($n = 7$), failure to save the EEG data due to hardware issues ($n = 1$), and poor EEG data quality ($n = 1$). Thus, data from 25 participants were included in the final analysis (15 females; mean age: 22.24 ± 3.83 SD years old). All subjects had a normal or corrected-to-normal vision and reported no neurological or psychiatric disorders. Participants provided written informed consent before participation. This study was conducted in accordance with a protocol approved by the Institutional Review Board of the Institute of Psychology, Chinese Academy of Sciences.

2.2 | Procedure

Participants completed an intentional encoding task and a 24-h delayed recognition test. We adopted the choice memory paradigm used in our previous research (Ding et al., 2021; Xue et al., 2023), which was modified from the paradigm used in studies by Murty et al. (2015, 2019). Three hundred and twenty-six images of outdoor and indoor scenes were used in our experiment. In the encoding phase, 166 trials/images in total were used, including 6 trials (3 indoor and 3 outdoor images) used for primacy and recency fillers (not included in the subsequent test and analyses) and 80 trials (40 indoor and 40 outdoor images) for each condition. In the test phase, 320 trials/images in total were presented, with 160 previously presented images and 160 new images.

During the encoding phase (see Figure 1a), each trial started with the presentation of a cue for 1.5 s that indicated the trial type (i.e., "choice [选择]" or "fixed [非选择]" condition), followed by a jitter fixation cross for 1.5–2.5 s. Then, two occluder screens of question marks in circles were displayed on the left and right sides. Participants were instructed to remove one of the occluder screens and were allowed 2 s to memorize the image underneath. In the choice condition, participants could make their own choice to remove either the left or right occluder, whereas in the fixed condition, participants were forced to remove the predetermined occluder on the side with a red arrow. In the current paradigm, the memorandum of each trial was predetermined and matched between the two conditions, so differences in memoranda should not influence memory performance and brain activity. After an approximately 24-h delay, participants returned for a recognition test (see Figure 1b), in which they were instructed first to judge whether the images were "Old" or "New" within 20 s and then to rate their level of confidence within 20 s with the following response options: "High Confidence," "Medium Confidence," and "Just Guessing" (2–3 s ITI).

2.3 | EEG acquisition

Participants sat approximately 60 cm from a display screen in a separate room from the experimenter. The experiment was programmed using PsychoPy and displayed on a screen with a resolution of 1280 × 1024 pixels and a refresh rate of 60 Hz. During the encoding phase, electrical brain activity was recorded using 64 channels (International 10-20 System of Electrode Placement) online referenced to FCz. Brain Products equipment was used for data acquisition. Eye movements and blinks were monitored by one additional electrode below the right eye (EOG). EEG signals were digitally sampled at 1000 Hz and amplified with a band-pass filter of 0.01 and 250 Hz. Impedances were kept below 10 kΩ.

2.4 | EEG preprocessing

EEG data were preprocessed offline using EEGLAB (Delorme & Makeig, 2004) and FieldTrip (Oostenveld et al., 2011) toolboxes in MATLAB 2019b. The online reference at the midline frontal site FCz was reinstated and used as a scalp site of interest. Two sites (FT9 and FT10)

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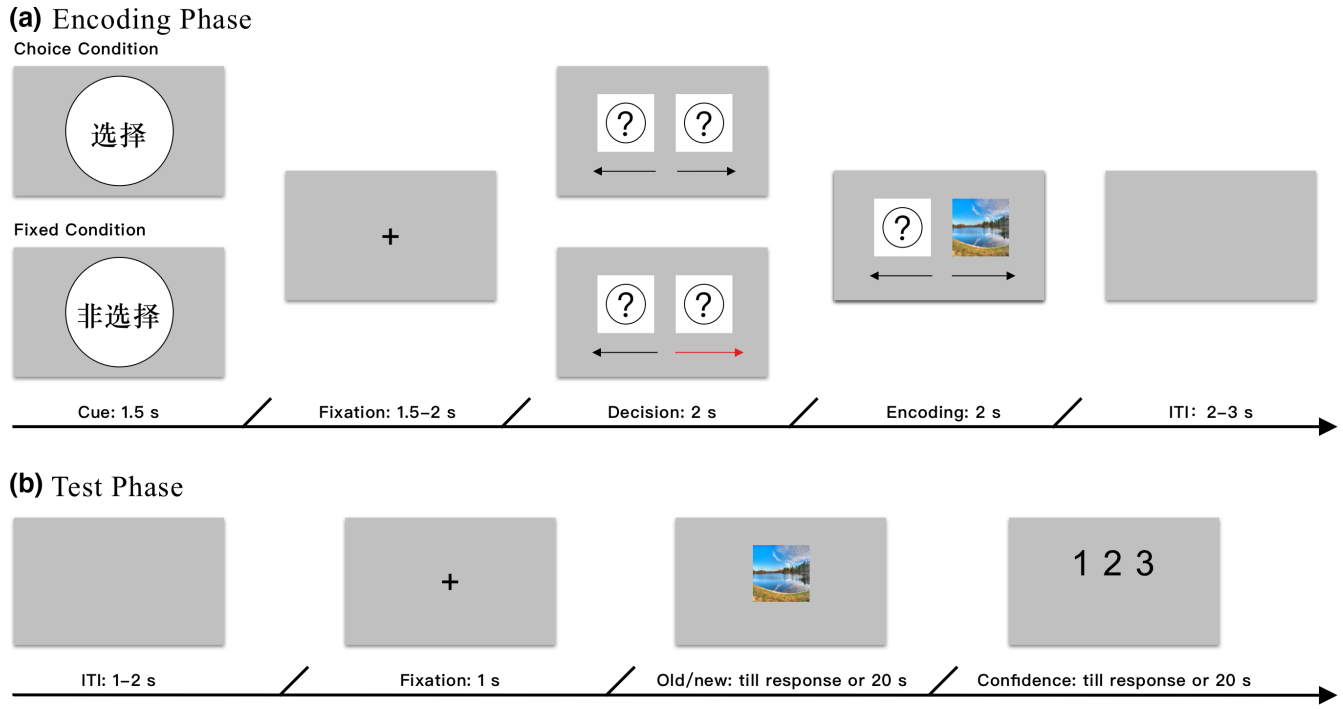


FIGURE 1 Schematic depiction of the choice encoding task and the test. (a) In the encoding phase, participants removed one of the occluder screens to memorize the item. In the choice (选择) condition, they could make their own choices to remove either the left or right occluder, whereas in the fixed (非选择) condition, they were forced to remove the one on the side with a red arrow. (b) In the test phase, participants were instructed to judge whether the images were "Old" or "New" and then to rate their level of confidence: 3 = "High Confidence," 2 = "Medium Confidence," and 1 = "Just Guessing."

were deleted and the continuous signals were re-referenced offline to the average of the left and right mastoids (TP9 and TP10). The data were resampled to 500 Hz and were filtered with a band-pass from 0.5 to 40 Hz. The continuous signals were divided into epochs from −2.5 to 2 s (0 s meant the time point of the picture onset; see the Time-frequency Analyses section for the reason that we chose the epoch of −2.5 to 2 s). They were baseline-corrected by using the mean signal of the time from −2.2 to −2 s (the last 0.2 s of the jitter fixation before the decision period). Epochs with a large burst of artifacts were deleted by a manual check. After interpolating bad channels, independent component analysis (ICA) was used to remove artifacts such as blinks, lateral eye movements, and voltage drifts by using the runica Infomax algorithm provided by EEGLAB (Delorme et al., 2007). Epochs with EEG voltages exceeding 100 μV were rejected.

2.5 | Time-frequency analyses

Time-frequency representations of powers were calculated using Morlet wavelets with 3 cycles in steps of 0.5 Hz from 1 to 40 Hz (Cohen, 2014). The relatively long epoch (−2.5 to 2 s) was used for time-frequency analyses to avoid edge effects. For each epoch, we removed 0.3 s at the edges (the beginning and the end), leaving an epoch between −2.2 to 1.7 s. The data outside of this time window were trimmed off. A baseline in the frequency domain was not employed, because we focused on the difference in oscillatory power between conditions (Fell & Axmacher, 2011; Guderian et al., 2009).

The time window of interest was −0.8 to 1.7 s, which was collapsed into fifty time bins of 50 ms each. The intervals of −0.8 to 0 s and 0–1.7 s were chosen as the pre-encoding and encoding periods of interest. We did not choose the whole 2 s of the decision period before the stimulus onset (the interval of −2 to 0 s) as the pre-encoding period of interest because our behavioral results showed the response for the decision was finished within 1.2 s for 99% trials and we excluded trials of response after the time point of 1.2 s, so that there would be no confounding influence of response on brain activity in the interval of −0.8 to 0 s. For each participant, EEG epochs were binned according to the condition (choice vs. fixed) and subsequent memory performance (remembered vs. forgotten). Trials with no response or incorrect response (i.e., in the fixed condition, participants were forced to remove the left occluder but responded to the right side) or reaction time longer than 1.2 s were excluded. Similar to the study by Gruber and his colleagues (Gruber et al., 2013), we considered the confidence rating and excluded trials with "Just Guessing,"

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leaving trials with "High Confidence" and "Medium Confidence" for analyses. In addition, participants with fewer than 12 trials in any condition (remembered and forgotten trials in the fixed and choice conditions) were excluded. Finally, the final analyses included on average 65.88 ± 10.69 trials in the choice condition (39.72 ± 12.50 remembered trials and 26.16 ± 10.39 forgotten trials) and 65.80 ± 11.80 trials in the fixed condition (38.68 ± 11.63 remembered trials and 27.12 ± 10.21 forgotten trials).

2.6 | Statistical analyses

Three sets of analyses were conducted with the EEG data. First, following the procedures used in previous studies (Gruber et al., 2013), we examined the interaction and main effects of condition (choice vs. fixed) and SME (remembered vs. forgotten) on the theta band (between 4 and 7 Hz; Hanslmayr et al., 2009) for each time point and each electrode, using a non-parametric cluster-based permutation approach with the FieldTrip toolbox (Blair & Karniski, 1993; Hanslmayr et al., 2009; Maris & Oostenveld, 2007). F-statistics were used to examine the interaction effect and the main effects of condition and SME separately by using contrasts of interest. Any significant interaction effect was followed up with simple effect analyses (i.e., SMEs in each condition). In the cluster-based permutation approach, adjacent electrodes showing significant differences between levels of a contrast ($p < .05$) were grouped into clusters. The sum of statistical values within each cluster was defined as the test statistic. Next, a null distribution of test statistic was created using a Monte Carlo approximation, which involved 1000 iterations of shuffling the contrast labels within subject. Finally, the clusters with an empirical test statistic larger than the 95th percentile of the null distribution ($p < .05$) were considered statistically significant. Only the effects that spanned across a minimum of 2 neighboring electrodes and four consecutive time bins (200 ms) were considered to be reliable and used for further analyses.

The second set of analyses were conducted to control for potential effects of participants' preference of the right or the left side when they made their choices in the choice condition (i.e., a participant might choose the items in one side more than the other side). Specifically, we extracted and averaged across all data points (at each electrode and each time point) that were significant for a given contrast and applied linear mixed models on the averaged data of each contrast with the side preference (i.e., ratio of the left side to the right side) as a covariate. Random subject intercepts were included in the mixed models and the *p*-values reported for the mixed models were calculated using the maximum likelihood estimation.

Finally, we conducted a set of analyses examining the alpha and beta bands. Although the focus of this study was on the theta band because of its close relation to motivation and memory (Begus & Bonawitz, 2020; Gruber et al., 2013; Pacheco Estefan et al., 2021), we explored the roles of the alpha and beta bands because they have been shown to be involved in memory (e.g., Hanslmayr et al., 2009; Hanslmayr et al., 2011; Hanslmayr et al., 2012). We also used the non-parametric cluster-based permutation approach as described above to analyze the data for the alpha (8–12 Hz), low beta (13–17 Hz), and high beta (18–30 Hz) bands. We applied Bonferroni corrections for multiple comparisons for the analyses of these frequency bands.

3 | RESULTS

3.1 | Behavioral results

Paired *t* tests were conducted to test the differences in several indices of memory performance in the recognition test between the choice and fixed conditions. Consistent with previous studies (Ding et al., 2021; DuBrow et al., 2019; Murty et al., 2015), the results showed the choice effect for accuracy (mean difference ± SD = 0.028 ± 0.057, $t_{(24)} = 2.463$, $p = .021$; see Figure 2). To consider false alarms, we calculated the d' score (the z score of the hit rate/accuracy minus the z score of the false alarm rate) and also detected the choice effect for d' score (mean difference ± SD = 0.079 ± 0.164, $t_{(24)} = 2.418$, $p = .024$). When

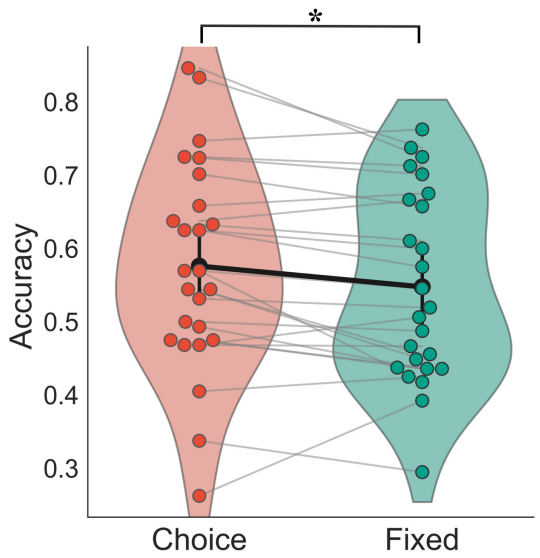


FIGURE 2 The choice effect on memory performance. The violin plot shows the accuracy rate by condition (choice vs. fixed), with each linked pair of dots representing the accuracy rates of a participant for the two conditions. *$p < .05$.

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the confidence ratings were considered (i.e., only trials with "High Confidence" and "Medium Confidence" were included in the analysis), the choice effects for accuracy and d' score were not statistically significant, but showed weak trends (accuracy with confidence: mean difference $\pm$ SD $= 0.021 \pm 0.066$, $t_{(24)} = 1.602$, $p = .122$, and d' score with confidence: mean difference $\pm$ SD $= 0.061 \pm 0.187$, $t_{(24)} = 1.624$, $p = .117$). We also conducted again the above t tests including the data of the nine participants (total $n = 34$) that were excluded from the EEG analyses (see the section of *Participants* for details). The results showed a similar pattern (see Supplementary Materials).

3.2 | Theta oscillatory power

First, we observed a cluster in the left frontal–parietal area (AF7, F7, F5, F3, FT7, FC5, FC3, FC1, T7, C5, and C3, see Figure 3a,b) that exhibited a significant interaction effect during the encoding period (from 0.15 to 0.4 s after stimulus onset) in the theta band (summed $F = 66.19$, $p = .048$), with greater SME in the choice condition than the fixed condition. Further simple effect analyses revealed that the interaction effect was caused by a significantly positive SME (larger theta power for subsequently remembered than forgotten items) in the left frontal–parietal area (F7, F5, F3, FT7, FC5, FC3, FC1, FCz, T7, C5, and C2, see Figure 3c) in the choice condition ($p = .046$), but not in the fixed condition (see Figure 3d).

Then, we identified a cluster in the parietal channels (F4, FC3, FC1, FCz, FC2, FC4, C1, Cz, C2, C4, CP1, CPz, CP2, and Pz) that exhibited a negative main effect of SME from 0.4 to 0.7 s after stimulus onset (less theta power in the subsequently remembered than forgotten trials, summed $F = 86.45$, $p = .036$, see Figure 4a). Moreover, we identified two clusters, each exhibiting a main effect of condition during the encoding period in the theta band (see Figure 4b). One cluster showed a positive main effect of condition (i.e., greater theta power in the choice than the fixed condition) from 0 to 0.3 s after item presentation, distributed in the left and right parietal-occipital areas (Cz, CP3, CP1, CPz, CP2, CP4, P3, P1, Pz, P2, P4, P6, and PO3, summed $F = 100.02$, $p = .035$). The other cluster displayed a negative main effect of condition (less theta power in the choice condition than the fixed condition) from 0.4 to 0.7 s after item presentation, distributed over the right frontal, temporal and occipital channels (Fp2, AF4, AF8, F6, F8, FC6, FT8, C6, T8, CP6, P6, P8, PO8, and O2, summed $F = 93.82$, $p = .026$).

As aforementioned, we also extracted and averaged data with each significant effect in theta band reported above and conducted analyses controlling for the difference in the number of chosen items between the two sides. After including the ratio of one side as a covariate, the results still showed significant interaction effect between the condition and SME during the time window of 0.15–0.4 s ($b = 0.16$, 95% CI [0.05, 0.27], $p = .005$), main effect of SME during the time window of 0.4–0.7 s ($b = 0.061$, 95% CI [0, 0.12], $p = .049$), and main effects of condition during the time window of 0–0.3 s ($b = 0.09$, 95% CI [0.03, 0.16], $p = .006$) and the time window of 0.4–0.7 s ($b = 0.07$, 95% CI [0.02, 0.12], $p = .011$).

In addition to the theta band, we explored the alpha and low beta bands and found a negative main effect of condition (less power in the choice condition than the fixed condition) in a cluster in the parietal area during the encoding period (alpha: from 0.6 to 0.9 s after stimulus onset, corrected $p = .042$; low beta: from 0.7 to 0.9 s after stimulus onset, corrected $p = .012$, see Figure 5e). All contrasts of interest across time points and frequencies can be seen in Figure 5. Further details regarding the uncorrected results for the alpha and beta bands are displayed in the Supplementary Materials.

4 | DISCUSSION

In the current study, we investigated the role of theta oscillations in the formation of choice-related memory. The participants in the current study performed an inconsequential choice memory task in which memoranda were selected either by themselves (the choice condition) or by the computer (the fixed condition). The two conditions were otherwise the same. Our behavioral data revealed that the participants retained better one-day delayed memory of objects in the choice than in the fixed condition. Our EEG data showed that increases in theta power during an early latency of encoding period (0.15–0.4 s) predicted subsequently memory performance in the choice condition, but not in the fixed condition, suggesting a specific role of theta power during the early latency in the choice-related memory formation. We also found that decreases in theta power during a late latency (0.4–0.7 s) predicted subsequent memory performance in both the choice and fixed conditions. In addition, we observed an increase in theta power in the choice condition compared to the fixed condition in the early latency (0–0.3 s) and a decrease in theta power in the choice condition compared to the fixed condition in the late latency (0.4–0.7 s). These results demonstrated the critical role of theta oscillations in choice-related memory formation, which would be discussed below in detail.

Our finding that theta power at left frontal–parietal sites following the stimulus predicted subsequent memory only in the choice context is consistent with the patterns of two previous studies on theta oscillations and

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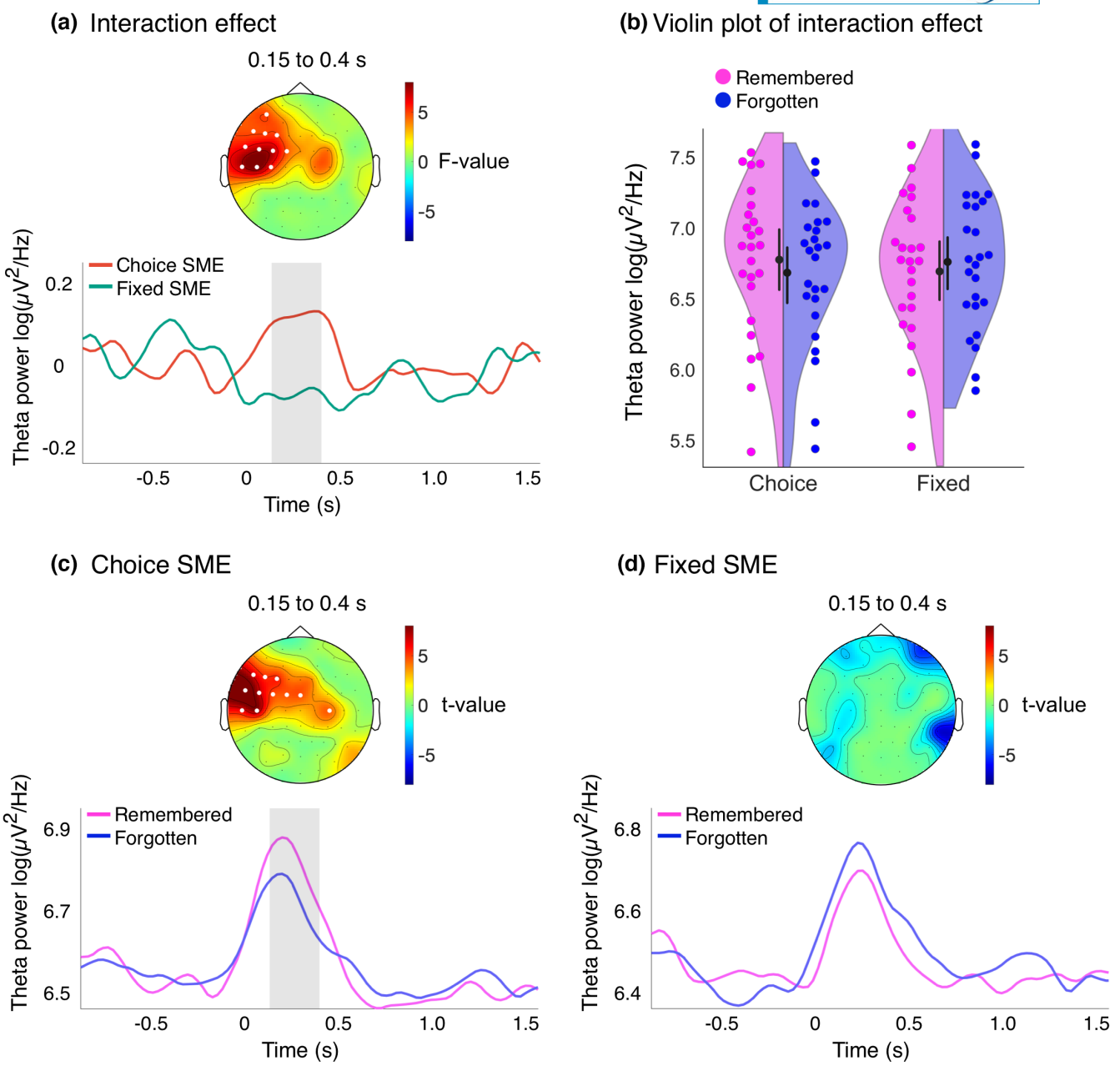


FIGURE 3 Interaction effect between condition and SME as well as the simple effect by conditions in the theta band. (a) The positive interaction effect with larger choice SME than fixed SME in the time window from 0.15 to 0.4 s. The topographical map shows the statistical distribution, with the significant cluster marked by white dots. The line plots depict the average time courses across channels within the cluster, with significant time points shaded in gray. (b) Violin plots of power amplitude (log transformed) for the cluster that showed significant interaction effect between condition and SME. Error bars represent ±SEM. (c) The positive SME in the choice condition with larger theta power for later remembered than forgotten items in the time window from 0.15 to 0.4 s. (d) No significant SME in the fixed condition in the time window from 0.15 to 0.4 s. The line plots depict the average time courses across channels within the cluster of significant SME in the choice condition.

monetary reward-related memory (Gruber et al., 2013; Pu & Yu, 2019). Both previous studies found a positive theta power SME specifically for the high reward or reward condition and concluded that monetary reward facilitated memory performance by inducing theta oscillatory activity. The self-determined choice itself is believed to be internally rewarding as it activates the dopaminergic reward system (i.e., the ventral striatum) just like the external reward that enhances behavioral performance (Murty et al., 2015, 2019). Taken together, these two studies and our current study, theta oscillations seem to play a general role in the formation of both external and internal reward-related memory. It is worth noting, however, that the three studies found

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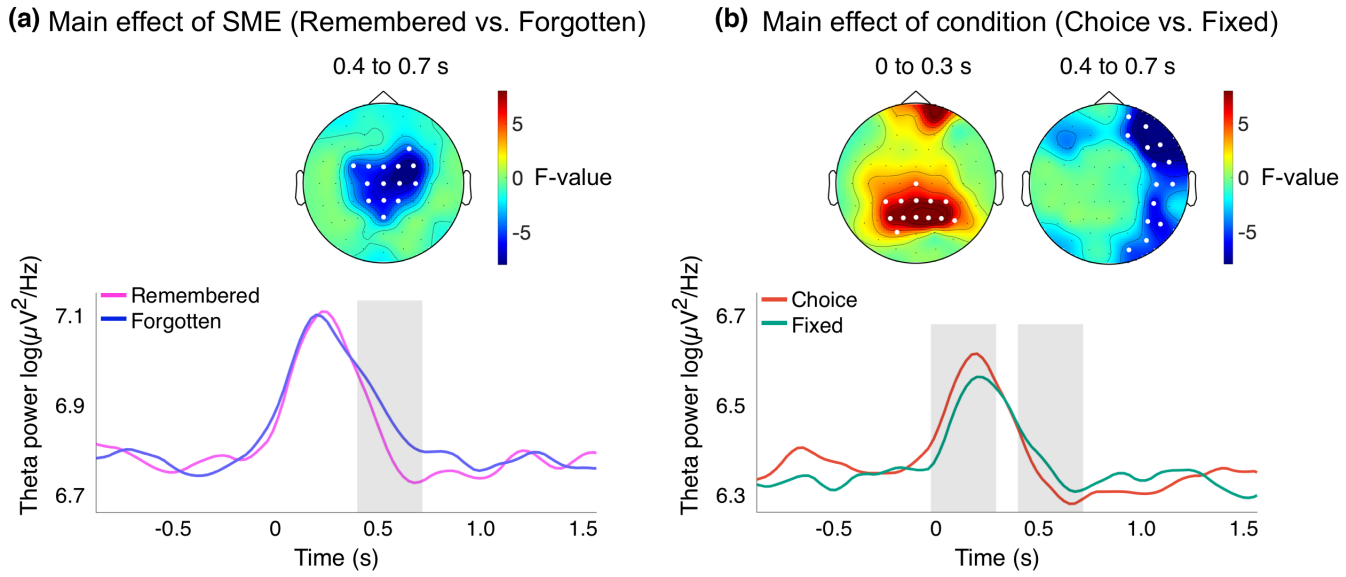


FIGURE 4 Main effects of SME and condition in the theta band. (a) The negative main effect of SME with larger theta power for later remembered than forgotten items in the time window from 0.4 to 0.7 s. The topographical map shows the statistical distribution, with the significant cluster marked by white dots. The line plots depict the average time courses across channels within the cluster, with significant time points shaded in gray. (b) The positive main effect of condition with larger theta power for items in the choice condition than in the fixed condition in the time window from 0 to 0.3 s and the negative main effect of condition in the time window from 0.4 to 0.7 s. The line plots depict the average time courses on the representative channels (CP4 and P4) within the two clusters.

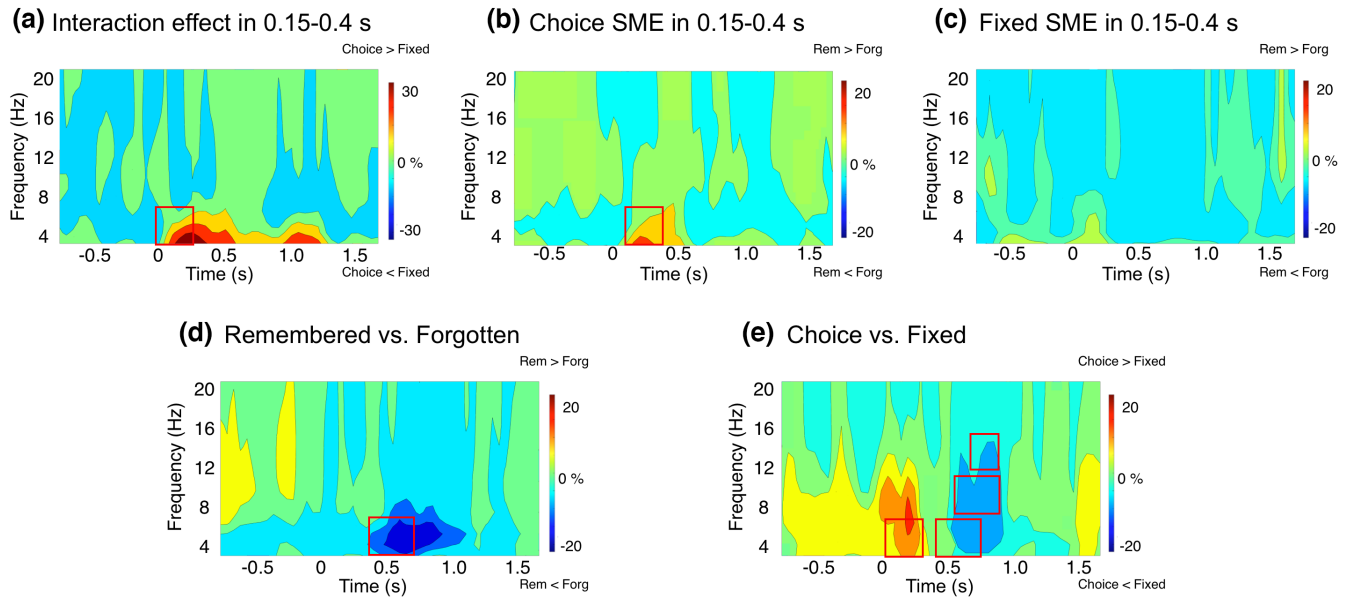


FIGURE 5 Time–frequency plots of all contrasts of interest averaged across representative electrodes. Note that these plots do not form the basis of our analysis as we focused on the predetermined theta band frequency. The plots presented here serve only as an illustration of the development of effects across the time and frequency domains. Red squares mark the time windows of the significant effects of power bands. (a) Interaction effect between condition (choice vs. fixed) and SME (remembered vs. forgotten). (b) Choice SME in 0.15 to 0.4 s. (c) Fixed SME in 0.15 to 0.4 s. (d) Main effect of SME (remembered vs. forgotten). (e) Main effect of condition (choice vs. fixed).

different time windows: the pre-encoding period in Gruber et al. (2013), the post-encoding period in Pu and Yu (2019), and the encoding period in our study. These differences in results may be due to task differences in the three studies and future studies need to replicate these results with varying tasks. As Shohamy and Adcock (2010) suggested, mesolimbic dopamine modulates memory over a range of timescales, depending on various factors such as salience of events and sustainability of contexts.

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What are the mechanisms underlying the role of theta oscillations in choice-related memory? Previous neuroimaging studies have demonstrated that having choices as well as other types of reward enhances memory performance through dopaminergic modulation of activity in the hippocampus (Gruber et al., 2014; Murty et al., 2015). The reward-induced engagement of the mesolimbic dopamine system can modulate hippocampal physiology and enhance synaptic plasticity that lead to successful episodic memory encoding (Lisman et al., 2011; Shohamy & Adcock, 2010). Critically, previous studies have revealed that theta oscillations induced by reward are localized in the dopamine reward circuitry of the brain (Cohen et al., 2012). In addition, it has also been found that theta activity can activate a mnemonic context, which allows the presented item to be better encoded, enhances synaptic plasticity in the hippocampus, and improves memory formation (Axmacher et al., 2006; Düzel et al., 2010; Guderian et al., 2009). Thus, it is possible that making choices as an intrinsic reward activates a mnemonic state through theta activity, which then indexes the interaction between the reward dopaminergic system and the hippocampus memory system during the encoding period. Indeed, we found the choice-related enhancement of theta power during the similar encoding period (0–0.3 s, increased theta power in the choice condition compared to the fixed condition). Consistent with our results, a recent iEEG study showed that when people can actively control the learning process, increased hippocampal theta oscillations during the encoding period was linked to enhanced memory performance (Pacheco Estefan et al., 2021). Together, these findings provide a convergence between two diverse lines of research – one relating theta oscillations to choice/self-determination, and the other linking theta oscillations to memory performance.

It should be noted that previous studies also linked theta oscillations to attention and cognitive effort (Bosseler et al., 2013) and top-down control for selective encoding of information (Nyhus & Curran, 2010). Gruber and his colleagues' finding of the increased theta activity during the pre-encoding period in the high reward condition might reflect the voluntary control for attention and cognitive effort to prepare for encoding (Gruber et al., 2013). However, attention and cognitive control would not be able to account for our results for two reasons. First, Murty et al.'s (2015) eye-tracking study ruled out the attentional account for choice-related memory enhancement. They found that looking times were greater in the fixed condition than in the choice condition. Second, the choice SME in our study did not occur before the encoding period.

In addition to the specific SME for the choice condition, we revealed a main effect of negative SME (0.4–0.7 s), suggesting that the decreased theta power worked as a common mechanism signaling the successful encoding of items. Although several previous EEG and MEG studies of humans have revealed negative SMEs, their interpretation remains speculative because these studies did not investigate these theta decreases in more detail (Burke et al., 2013; Fell et al., 2008; Guderian et al., 2009; Lega et al., 2012; Sederberg et al., 2007). In contrast, previous iEEG studies of animals revealed specific mechanisms. They detected decreased theta power during active behavior (Poulet et al., 2012; Poulet & Petersen, 2008) and traced them to a reduced correlation between individual neurons, which might enable more efficient local computation (Harris & Thiele, 2011). In other words, the decreases in theta power might be a mechanism to de-correlate neural activity to enhance the neural coding capacity and to promote memory performance (Hanslmayr et al., 2016). Interestingly, we also detected choice-related decrease in theta power in the same encoding period (0.4–0.7 s, decreased theta power in the choice condition compared to the fixed condition). It is possible that making choice would trigger neurons' more efficient responses that could benefit memory formation. The similar pattern of encoding- and choice-related activity during the later latency as well as during the early latency as aforementioned may suggest an interplay between choice and encoding in the service of the presented object (Adcock et al., 2006; Gruber & Otten, 2010; Wittmann et al., 2007).

In addition to the results related to the theta band, our exploratory broadband analysis revealed decreased alpha/beta power (alpha/beta suppression) in the choice condition compared to the fixed condition (alpha: 0.6–0.9 s; low beta: 0.7–0.9 s). Moreover, we observed increased alpha/beta suppression for successful encoding in the choice condition (see the uncorrected results in the Supplementary Materials). Alpha/beta suppression has been linked to an active brain state that is beneficial to task performance (Gable et al., 2016; Glazer et al., 2018; Meyniel & Pessiglione, 2014). It has also been associated with successful encoding, perhaps reflecting active information processing or rich information representation (Griffiths et al., 2019; Hanslmayr et al., 2009; Hanslmayr et al., 2012; Hanslmayr et al., 2016; Klimesch et al., 1996; Sederberg et al., 2003). Taken these results together with our finding, it seems possible that the choice condition in the current study reduced alpha/beta synchrony, which led to an active brain state that is conducive to memory encoding.

Finally, we should comment on the nonsignificant (weak trend) choice effects after controlling for confidence ratings. In our previous behavioral study using a similar paradigm (Xue et al., 2023), we observed significant choice effects after controlling for confidence ratings. It turns out that the effect size in the current

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study ($d = 0.32$) is almost the same as that of our previous work ($d = 0.31$). Therefore, the nonsignificance of this study's choice effect after controlling for confident ratings seems to be due to its smaller samples size (25 participants in the current study as compared to 36 participants in Xue et al.' study). Indeed, when we included the data of the nine participants who were excluded from the EEG analyses, we detected a marginally significant choice effect for the accuracy ($p = .055$) and a significant effect ($p < .05$) for the d' score ($n = 34$, see the Supplementary Materials for details) even after controlling for confidence ratings. Future EEG studies should increase sample size to obtain a stronger statistical power. Another methodological note is that the fixed condition included a red arrow, which introduced a difference in visual input between conditions. This factor could have potentially influenced the results and should be controlled for in future studies.

In conclusion, this study revealed the role of theta oscillations in enhancing episodic memory when individuals had an opportunity to choose which items to memorize. These findings contribute to our understanding of the mechanisms underlying active and motivational learning by linking theta activity to choice and memory formation.

AUTHOR CONTRIBUTIONS

Si Cheng: Data curation; formal analysis; methodology; visualization; writing – original draft. **Zhuolei Ding:** Data curation; formal analysis; project administration; visualization; writing – original draft. **Chuansheng Chen:** Conceptualization; writing – review and editing. **Wenxiang Sun:** Data curation; project administration. **Ting Jiang:** Writing – review and editing. **Xun Liu:** Funding acquisition; supervision; writing – review and editing. **Mingxia Zhang:** Conceptualization; funding acquisition; resources; supervision; writing – review and editing.

ACKNOWLEDGMENTS

This research was funded by the Natrual Science Foundation of China (62177045, 31500888), the German Research Foundation (DFG) and the National Science Foundation of China (NSFC) in project Crossmodal Learning (DFG TRR-169 / NSFC 61621136008).

CONFLICT OF INTEREST STATEMENT

The authors declare no competing financial interests.

DATA AVAILABILITY STATEMENT

The data and analysis code that support the findings of this study will be made available from the author, Si Cheng (chengsi123456@gmail.com), upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Main effects of condition in the alpha band (8–12 Hz). Two clusters showed significant main effect of condition (choice vs. fixed): one cluster from 0 to 0.3 s after stimulus onset showed positive effect and the other cluster from 0.6 to 0.9 s after stimulus onset showed negative effect. The topographical maps depict the statistical distributions, with the significant clusters marked by white dots. The line plot depicts the time course of the alpha power on channels Cz, CPz, and Pz. Significant time points are shaded gray.

Figure S2. Main effects of condition in the low beta band (13–17 Hz). One cluster from 0.7 to 0.9 s after stimulus onset showed negative main effect of condition. The topographical maps depict the statistical distributions, with the significant clusters marked by white dots. The line plot depicts the time course of the low beta power on channels Cz, CPz, and Pz. Significant time points are shaded gray.

Figure S3. Interaction effect between condition and SME in the low beta band (13–17 Hz). (A) A negative interaction effect in the time window from 0 to 0.4 s. The topographical map showed the statistical distributions of the interaction effect, with the significant cluster marked by white dots. The line plot depicts the time course of the choice SME versus the fixed SME on channels of C5, C3, C1, CP5, CP3, and CP1. Significant time points are shaded gray. (B) Violin plots of power amplitude (log transformed) for the cluster that showed significant interaction effect between conditions and SME. Error bars represent ±SEM. (C) The negative SME with less low beta power for later remembered than forgotten items in the time window from 0 to 0.4 s in the choice condition. The topographical map showed the statistical distributions, with the significant cluster marked by white dots. The line plot depicts the time course of the choice SME on channels C5, C3, C1, CP5, CP3, and CP1. Significant time points are shaded gray. (D) No significant SME in the time window from 0 to 0.4 s in the fixed condition. The topographical map showed the statistical distributions. The line plot depicts the time course of the fixed SME on channels C5, C3, C1, CP5, CP3, and CP1.

Figure S4. Interaction effect between condition and SME in the high beta band (18–30 Hz). (A) A negative interaction effect with less choice SME than fixed SME in the time window from 0 to 0.3 s. The topographical map showed the statistical distributions of the interaction effect, with the significant clusters marked by white dots. The line plot depicts the time course of the choice SME versus the fixed SME on channels C5, C3, C1, CP5, CP3, and CP1. Significant time points are shaded gray. (B) Violin plots of power amplitude (log transformed) for the cluster that showed significant interaction effect between condition and SME. Error bars represent ±SEM. (C) The negative SME with less high beta power for later remembered than forgotten items in the time window from 0 to 0.3 s in the choice condition. The topographical map showed the statistical distributions, with the significant cluster marked by white dots. The line plot depicts the time course of the choice SME on channels C5, C3, C1, CP5, CP3, and CP1. Significant time points are shaded gray. (D) No significant SME in the time window from 0 to 0.3 s in the fixed condition. The topographical map showed the statistical distributions. The line plot depicts the time course of the fixed SME on channels C5, C3, C1, CP5, CP3, and CP1.

How to cite this article: Cheng, S., Ding, Z., Chen, C., Sun, W., Jiang, T., Liu, X., & Zhang, M. (2023). The effect of choice on memory: The role of theta oscillations. *Psychophysiology*, *60*, e14390. https://doi.org/10.1111/psyp.14390

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